Allen Saji

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SUMMARY

Final-year engineering student at QUT with a hybrid software and electrical background. Project experience spans AWS cloud architecture, React frontend development, embedded IoT systems, power electronics, computer vision research, and leading a five-person drone payload team as project manager. Open to internships and graduate roles in software or electrical engineering.

EDUCATION

Queensland University of Technology Expected Early 2027
Bachelor of Engineering (Honours), Computer and Software Systems Brisbane, QLD

- Minoring in Advanced Electrical Engineering and Robotics & Embedded Systems
- Dean's List

Indian School Al Ghubra Jun. 2022
High School Diploma – 7.0 GPA Muscat, Oman

- Executive, Robotics Club (2018 – 2020)

PROJECTS

Honours Research: Bubble Analysis in Water Electrolysers | *Python, PyTorch, Detectron2* Feb. 2026 – Oct. 2026

- Built a computer vision pipeline that automatically outlines bubbles in 2,000 fps electrolyser video, training Mask R-CNN on a dataset of 37,388 labels audited from three annotators' work
- Improved segmentation accuracy (mask AP) from 40.4 to 45.1 by fine-tuning at a higher input resolution, reaching an F1 of 0.80, and compared the model with StarDist and a hybrid on held-out test data
- Detected 94–100% of bubbles above 25 px, including overlapping ones, and measured gas coverage and bubble size (Sauter mean diameter) to within about 3%; total counts ran about 20% low because very small bubbles were missed
- Measured how often human labellers agree and found the model's small-bubble limit (under about 15 px) lines up with where they stop agreeing, making a consistent labelling rule the top recommendation

UAV Air Quality and Target Detection Payload | *Raspberry Pi 5, Python, MQTT, I2C/SPI* Jul. 2026 – Oct. 2026

- Project manager and air quality lead for a five-person team building a 364 g drone payload that monitors air quality, reads pressure gauges and drills soil samples
- Led systems engineering across four subsystems, defining the interfaces and communication standards between them and verifying the integrated payload against the customer's requirements
- Wrote the embedded Python sensor software on a Raspberry Pi 5, sampling temperature, humidity, pressure, light and gas over I2C and correcting temperature for CPU heat
- Streamed telemetry over MQTT to the ground station dashboard, and built an SPI LCD controller showing live sensor data and detection images, switched remotely from the dashboard

Cloud-Based Background Removal System | *AWS, Node.js, Docker* Aug. 2025 – Nov. 2025

- Designed and deployed a cloud-native background removal platform on AWS as two independent containerised services: a stateless API tier and an ML processing tier, orchestrated with ECS Fargate for serverless compute and auto-scaling
- Architected an asynchronous SQS pipeline to decouple user requests from ML inference, with retry logic and a Dead Letter Queue to automatically isolate failing jobs after three attempts
- Integrated a neural network-based background removal model into the processor service, with processed images stored in S3 and job status tracked through PostgreSQL on RDS
- Secured the platform with AWS Cognito authentication and S3 pre-signed URLs so image data bypassed the API servers entirely, keeping the compute tier stateless and horizontally scalable; deployed behind an ALB with HTTPS and a custom domain

Portfolio Website | *React 19, Vite, three.js, CSS* Dec. 2025 – Present

- Designed and built allensaji.com end-to-end around a scroll-driven 3D descent from deep space to a moon, rendered in three.js with custom flat-shaded, dithered shaders and a camera path tied to scroll position
- Generated the whole scene in code, with no imported models or textures: planets, rings, constellations and a rocket with a camera-facing trail, keeping camera and flight logic in pure functions covered by over 170 Vitest tests
- Fed the descent and a compact portfolio from one shared content source, with phone layouts, light and dark themes, reduced-motion and no-WebGL fallbacks, and keyboard navigation; deployed on Vercel

Solar Inverter Reverse Engineering | *LTspice, MATLAB* Feb. 2025 – Jun. 2025

- Reverse-engineered a commercial 5 kW grid-tied PV inverter with a partner, reconstructing both the DC-DC boost stage and single-phase full-bridge DC-AC stage from physical measurements and datasheet specifications
- Modelled the boost converter in LTspice and MATLAB, validating theoretical ripple and duty cycle calculations against simulation and estimating stage efficiency after accounting for inductor, switching, and capacitor losses
- Designed the full-bridge inverter stage using sinusoidal pulse-width modulation to push switching noise above the audible range, selecting a suitably rated IGBT and sizing the output filter to meet grid voltage and waveform quality requirements
- Proposed a snubber circuit to significantly reduce switching losses in the boost MOSFET, and identified synchronous rectification as a further efficiency improvement path

EXPERIENCE

Coles May 2024 – Present
Team Member (Service and Online) Brisbane, QLD

- Led day-to-day operations within the online team, coordinating workflows to ensure efficient order fulfilment and high customer satisfaction
- Acted as a first point of contact for customer issues, resolving disputes and escalating when necessary to maintain service quality
- Trained and mentored new team members, improving team performance, consistency, and adherence to procedures
- Managed high-volume transactions using point-of-sale systems with accuracy and efficiency, maintaining smooth store operations

Wildfire Sports & Trek Oct. 2023 – Feb. 2024
Warehouse Dispatcher Brisbane, QLD

- Reduced order processing times with streamlined picking, packing, and shipping procedures
- Participated in continuous improvement initiatives to streamline workflows and optimise productivity
- Prepared shipments for safe transportation by securely wrapping parcels and labelling packages according to company and shipping guidelines

SKILLS

Programming : Python, C, C#, JavaScript/Node.js, HTML/CSS
Frontend : React, Vite, three.js, Framer Motion
Cloud : AWS (ECS Fargate, EC2, SQS, S3, RDS, Cognito, ALB, CloudFront), Docker, Git, Linux (WSL)
Databases : PostgreSQL, REST APIs
Engineering & Simulation : MATLAB/Simulink, LTspice, KiCad/Eagle, AutoCAD
Electrical & Power Systems : DC-DC converters, PV inverter modelling, PCB design & layout, circuit debugging
Embedded Systems : Raspberry Pi, microcontroller programming (C), MQTT, UART/SPI/I2C, sensor integration, hardware debugging
Machine Learning & Computer Vision : PyTorch, Detectron2, OpenCV, TensorFlow/Keras, image processing, instance segmentation, neural network integration
Languages : English (Native), Malayalam (Native)